

UX CASE STUDY · ENTERPRISE K-12 PLATFORM

Accessibility Program Architecture

*From Silence to System:
Architecting Accessibility at Scale*

PROGRAM TYPE

Accessibility Architecture

SCOPE

Enterprise K-12

YEAR

2024

PORTFOLIO CASE STUDY

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When Responsibility Became Infrastructure

The moment a practitioner becomes an architect

01

THE ORIGIN

In 2024, accessibility was not yet a structured initiative inside the organization. Federal regulatory pressure had increased urgency around compliance — but internally, accessibility was still treated as something reactive. Teams viewed it as a final-stage checklist item rather than a foundational part of product development.

At the time, I was the only accessibility subject matter expert across a department of more than 200 people.

I began as a practitioner — auditing interfaces, documenting issues, collaborating with developers, and validating fixes manually. But the scale of the problem exceeded what any single contributor could sustain. The product ecosystem spanned thousands of screens, multiple platforms, and highly complex workflows used by students, parents, teachers, and district administrators.

The challenge was no longer simply fixing accessibility issues. The challenge was creating a system capable of preventing them.

ROLE & SCOPE

This project became the turning point where my role evolved from UX practitioner into accessibility program architect — building organizational capability, operational systems, shared standards, and long-term cultural adoption.

HERO METRICS

ACCESSIBILITY CONTRIBUTORS

1 → 25+

From sole SME to cross-functional program

DESIGNERS TRAINED

7

First cohort; became internal advocates

ORGANIZATIONAL REACH

UX · QA · Engineering · Product

Full cross-functional adoption

PROGRAM TYPE

Accessibility Program Architecture

Systemic infrastructure, not point fixes

SCOPE

Enterprise K-12 Platform

Thousands of screens across multiple platforms

Understanding the Gaps No One Could See

Experiencing failure before designing solutions

02

PRODUCT ACCESSIBILITY ISSUES

Before defining solutions, I needed to understand where the failures truly existed. I began by experiencing the product through assistive technologies — screen readers, keyboard navigation, zoom testing, and color contrast validation.

The experience was fragmented and often unusable. Forms lacked accessible names. Navigation patterns broke keyboard flow. Semantic structure was inconsistent. Critical workflows could not be completed independently by users relying on assistive technology.

ORGANISATIONAL & CULTURAL GAPS

I immersed myself in the organization — observing design critiques, engineering workflows, QA processes, and product discussions. What I found was not active resistance, but a widespread absence of shared understanding.

Designers	Understood visual systems but not semantic structure
Engineers	Implemented patterns without understanding assistive technology behavior
QA Teams	Lacked accessibility validation criteria entirely
Leadership	Viewed accessibility as compliance, not design quality

To bridge these gaps, I facilitated accessibility exposure sessions across departments — demonstrating broken screen reader experiences live. Accessibility stopped being theoretical. It became tangible.

ATOMIC ACCESSIBILITY RESEARCH FRAMEWORK

Turning Testing Into Decisions

I adapted the Atomic Research model into an accessibility-specific framework. Rather than presenting disconnected accessibility bugs, findings were structured through four progressive layers — transforming

accessibility from subjective feedback into actionable organizational decision-making.

EXPERIMENT	FACT	INSIGHT	RECOMMENDATION
We tested... Screen reader, keyboard, zoom, voice control	We observed... Measurable interaction failures with % data	This suggests... User impact and behavioral implications	We should... Clear, prioritized remediation actions

ACCESSIBILITY TESTING INPUTS

Manual Keyboard Testing Validate complete non-mouse navigation across all flows
Screen Reader Testing Evaluate semantic interpretation and announcement accuracy
Voice Control Testing Assess command discoverability and label clarity
Color Contrast Validation Ensure visual readability across all states
Zoom / Magnification Testing Validate responsive scaling at 200–400%
Automated Scanning Detect technical violations at scale across components

ATOMIC RESEARCH EXAMPLES

EXPERIMENT Screen reader testing on forms	FACT 63% of inputs lacked accessible names	INSIGHT Users cannot complete forms confidently or independently	RECOMMENDATION Standardise accessible naming patterns across all inputs
EXPERIMENT Navigation usability testing	FACT 20% of users failed to locate key tasks	INSIGHT Navigation hierarchy lacks sufficient clarity	RECOMMENDATION Redesign navigation structure with semantic headings
EXPERIMENT A/B testing labels + icons	FACT 40% increase in task engagement with icons	INSIGHT Icons meaningfully improve comprehension	RECOMMENDATION Expand icon-supported labelling across the system

Reframing Accessibility as a Systems Problem

Shifting from screens to systems

03

THE FOUNDATIONAL INSIGHT

Once I understood the scale of the problem, I shifted from thinking about screens to thinking about systems. The platform contained thousands of interfaces, making one-off remediation impossible at any meaningful pace.



If we remediate the ~60 core components, we can influence thousands of downstream experiences simultaneously.

I mapped the product into its underlying component architecture, categorising issues across inputs and forms, navigation systems, data tables, validation patterns, structural hierarchy, and interactive states.

COMPONENT ECOSYSTEM MAP



Accessibility remediation at the component level cascades upward through templates, screens, and ultimately user journeys — creating leverage that manual remediation could never achieve.

ACCESSIBILITY RESPONSIBILITY MATRIX

Alongside the component architecture, I defined the organizational infrastructure required to sustain accessibility long term. Accessibility stopped being a specialised initiative and became operational infrastructure.

UX Design	Accessibility-first component design, semantic annotation, checklist adoption
Engineering	Semantic implementation, ARIA patterns, keyboard interaction models
QA	Accessibility acceptance criteria, screen reader validation, regression testing

Product	Accessibility in definition of done, roadmap prioritisation, SDLC integration
Leadership	Reporting structures, compliance accountability, culture investment

Scaling Through Capability Instead of Control

From dependency to distributed ownership

04

THE STRATEGIC SHIFT

Initially, I personally handled most remediation guidance. But centralising expertise created a bottleneck. If accessibility relied solely on me, it would never scale.



Rather than scaling output, I focused on scaling capability. The goal: build a system that teaches people to fish.

I designed a UX-centred accessibility training model tailored to the organisation's workflows and design systems. The first cohort included seven designers who began with foundational accessibility education before participating in collaborative remediation workshops directly.

These workshops were intentionally hands-on. We reviewed live product components together, identified failures, discussed semantic intent, and explored implementation considerations collaboratively. The process revealed deeper organisational gaps — especially between design intent and engineering execution.

Those early participants became advocates themselves, accelerating broader organisational adoption.

TRAINING EVOLUTION TIMELINE

P1

AUDIENCE

UX Designers

OUTCOME

Accessibility awareness and foundational knowledge

P2

AUDIENCE

UX + Engineering + QA

OUTCOME

Shared standards and cross-functional language

P3

AUDIENCE

Organisation-wide

OUTCOME

Embedded accessibility culture and self-sufficiency

Building the Infrastructure for Scale

The program itself was the prototype

05

WHAT WAS BUILT

The prototype in this project was not a screen. It was the program itself — a suite of tools, systems, and documentation structures designed to outlast any single contributor.

Accessibility Design Checklists

Reframed around UX workflows, not WCAG terminology

Annotated Figma Examples

Pattern library of accessible implementations with annotations

Accessibility Documentation Hubs

Centralised shared knowledge, searchable by stage and component

Testing Workflows

Step-by-step QA processes integrated into sprint ceremonies

Training Materials

Role-specific modules for Design, Engineering, and QA

Monthly Accessibility Newsletters

Keeping accessibility visible across the organisation

Shared Remediation Standards

Common language and criteria used across all disciplines

ACCESSIBILITY CHECKLIST DESIGN

Rather than organising guidance around WCAG terminology, the checklist was reframed around actual UX workflows and deliverables — making accessibility a natural part of existing design behaviors rather than a layer added afterward.

Exploration

→ Content hierarchy, semantic structure, heading levels, landmark regions

Refinement

→ Interaction states, focus management, accessible naming, labels

Handoff

→ Keyboard behavior, ARIA annotations, engineering implementation notes

LEADERSHIP RECOGNITION



Lacy demonstrated Lead UX-level ownership by defining and driving adoption of an Accessibility Design Checklist that helps designers consistently address accessibility earlier in the UX lifecycle. She showed cross-functional influence, systems-level thinking, and long-term organisational capability building.

— UX Leadership feedback, reflecting cross-functional influence and systems-level impact

Embedding Accessibility Into Everyday Delivery

From programme adoption to cultural standard

06

BEHAVIOUR CHANGE ACROSS TEAMS

As the program matured, accessibility became embedded into daily operations — not as a separate workstream but as an integrated quality standard across every discipline.

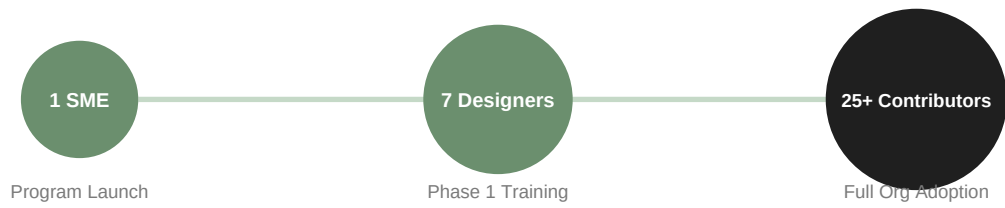
QA Teams	Adopted accessibility acceptance criteria as standard sprint gates
Designers	Began proactively identifying concerns before development began
Engineers	Started considering semantic implementation in early technical planning

THE LANGUAGE INSIDE THE ORGANISATION CHANGED

✕	"Does this look right?"
✓	"Is this keyboard navigable?"
✓	"Does this have an accessible name?"
✓	"Will a screen reader announce this correctly?"

That shift represented true cultural adoption. Accessibility had moved from external requirement to shared design quality standard.

ADOPTION GROWTH



Growth from one isolated subject matter expert to 25+ active contributors spanning UX, Engineering, QA, and Product.

Designing Systems That Continue Without Me

Sustainability as the true measure of impact

07

SUMMARY OF IMPACT

Within two years, accessibility evolved from a fragmented responsibility into a scalable organisational capability. The outcomes below represent systemic change — not individual heroics.

ACCESSIBILITY INTEGRATED INTO SDLC

Built into sprint ceremonies, not bolted on at the end

25+ TRAINED CONTRIBUTORS

Across UX, Engineering, QA, and Product functions

STANDARDISED REMEDIATION FRAMEWORKS

Consistent criteria used organisation-wide

SHARED KNOWLEDGE SYSTEMS

Documentation hubs that remain operational without ongoing maintenance

DESIGN SYSTEM GOVERNANCE

Accessibility-aware component standards embedded at the source



Most importantly, the system became sustainable. If I left tomorrow — the training, documentation, standards, and workflows would all continue.

KEY LEARNINGS



Scale through systems, not heroics — One expert cannot audit, fix, and maintain an enterprise product indefinitely. Systems, standards, and shared language are the only paths to scale.



Empathy precedes adoption — Live demonstrations of broken screen reader experiences changed minds faster than compliance arguments ever could.



Reframe around workflows, not regulations — Accessibility checklists organized by design stage — not WCAG criteria — achieved adoption because they matched how designers already work.



Build advocates, not dependencies — Training early participants to become internal advocates created an organic multiplication effect that no centralised program could replicate.



Sustainability is the final deliverable — The measure of program success is not what exists while you're there — it's what continues when you're gone.

Accessibility at scale is not achieved through expertise alone. It is achieved by building systems, shared language, and organisational capability that allow inclusive design to continue beyond any one individual.
